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1 Template

```

1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 #define all(v) v.begin(),v.end()
6 #define rall(v) v.rbegin(),v.rend()
7 #define pb push_back
8 #define sz(a) int(a.size())
9 #define F first
10 #define S second
11 typedef long long ll;
12 typedef vector<int> vi;
13 typedef vector<ll> vll;
14 typedef pair<int,int> pii;
15 typedef pair<ll,ll> pll;
16 const int MOD = 1e9+7;
17 const int INF = INT_MAX;
18 const ll INF64 = LLONG_MAX;
19 const long double EPS = 1e-9;
20 const long double PI = acosl(-1.0L);
21 void setIO(string p){
22     freopen((p + ".in").c_str(), "r", stdin);
23     freopen((p + ".out").c_str(), "w", stdout);
24 }
25
26 void solve(){
27
28 }
29
30
31 int main(){
32     ios_base::sync_with_stdio(0);
33     cin.tie(0);
34     cout.tie(0);
35     int tc = 1;
36     //cin >> tc;
37     for (int t = 1; t <= tc; t++)solve();
38 }

```

2 Data structures

2.1 Segment Tree

```

1 struct Node{
2     ll val = 0;
3 };
4 Node operator+(Node a,Node b){return {a.val+b.val};}
5 class Segment_Tree{
6 public:
7     int n;
8     vector<Node> tree;
9
10    Segment_Tree(int x = 1e5+10){
11        n = x;
12        tree.resize(2*n+1);
13    }
14
15    void update(int p, ll val){
16        for(tree[p+=n] = {val};p>1;p>>=1)
17            tree[p>>1] = tree[p]+tree[p^1];
18        //for (tree[p+=n] = {val};p>>= 1;)t[p] = t[p<<1]+t[p<<1|1];
19    }
20    ll query(int l,int r){
21        Node s;
22        for(l+=n,r+=n ; l<r ; l>>=1 , r>>=1){
23            if(l&1)s = s+ tree[l++];
24            if(r&1)s = s+ tree[--r];
25        }
26        return s.val;
27    }
28 };

```

2.2 Merge Sort Tree

```

1 struct Node{
2     vi val;
3
4     int search(int x){
5         int l = 0, r = sz(val)-1;
6         while(l<=r){
7             int m = l+(r-l)/2;
8             if(val[m] < x) l =m+1;
9             else r = m-1;
10        }
11        return sz(val)-1;
12    }
13 };
14
15 Node operator+(Node a,Node b){
16     int i=0,j=0;
17     Node aux;
18     while (i< sz(a.val) && j<sz(b.val)){
19         if (a.val[i]< b.val[j] ) aux.val.pb(a.val[i++]);
20         else aux.val.pb(b.val[j++]);
21     }
22     while (i<sz(a.val)) aux.val.pb(a.val[i++]);
23     while (j<sz(b.val)) aux.val.pb(b.val[j++]);
24     return aux;
25 }
26 //Call segment Tree

```

2.3 Fenwick Tree

```

1 struct Node{
2     ll n = 0;
3     Node operator+(Node b){return {n+b.n};}
4 };
5
6 class Fenwick_Tree{
7 public:
8     vector<Node> tree;
9     int n;
10    Fenwick_Tree(int x){
11        n = x+1;tree.resize(n);
12    }
13
14    ll sum(int r){
15        r++;
16        Node ans = {0};
17        while(r){
18            ans = ans + tree[r];
19            r -= r&-r;
20        }
21        return ans.n;
22    }
23
24    void update(int id, Node val){
25        id++;
26        while(id < n){
27            tree[id] = tree[id]+val;
28            id += id&-id;
29        }
30    }
31 };

```

2.4 Sparse Table

```

1 struct Node{
2     int val = 0;
3     Node operator+(Node b){return {__gcd(val,b.val)};}
4 };
5
6 class SparseTable{
7     public:
8     vector<vector<Node>> sparse;
9
10    SparseTable(int n=2e5+10){
11        sparse.assign(n , vector<Node> (log2(n)+1));
12    }
13
14    void build(const vi &a){
15        for(int i=0;i<sz(a);i++)
16            sparse[i][0] = {a[i]};
17
18        for(int j=1;(1<<j)<=sz(a);j++)
19            for(int i=0;i+(1<<j)-1 <sz(a);i++)
20                sparse[i][j]= sparse[i][j-1]+sparse[i+(1<<(j-1))][j-1];
21    }
22
23    int query(int l, int r){
24        int len = r-l+1;
25        int k = log2(len);
26        return (sparse[l][k]+sparse[r-(1<<k)+1][k]).val;
27    }
28
29 };

```

2.5 Disjoint Set Union

```

1 class DSU{
2     private:
3     vi parent,rank,size;
4     int c;
5     public:
6     int mx = 0;
7     DSU(int n):parent(n+1),rank(n+1,0),size(n+1,1),c(n){
8         iota(all(parent),0);
9     }
10
11    int find(int u){return parent[u] == u?u:parent[u] = find(parent[u]) ;}
12
13    bool same(int u,int v){return find(u)==find(v);}
14
15    int get_size(int u){return size[find(u)];}
16
17    int count(){return c;}
18
19    void merge(int u,int v){
20        u = find(u);
21        v = find(v);
22        if(u!=v){
23            c--;
24            if(rank[u] > rank[v])swap(u,v);
25            parent[u] = v;
26            size[v] += size[u];
27            if(rank[u] == rank[v])rank[v]++;
28            mx = max(mx,size[v]);
29        }
30    }
31 };

```

2.6 Lazy SegTree

```

1 struct Segment_Tree{
2
3     int n;
4     vector<Node> tree;
5     vector<Lazy> lazy;
6
7     Segment_Tree(int x = 1e5+10){
8         n = x;
9         tree.resize(4*n);
10        lazy.resize(4*n);
11    }
12
13    void push_set(int v,int l,int r){
14        if(!lazy[v].has_set)return;
15        tree[v].val = lazy[v].val*(r-l+1);
16
17        if(l!=r){
18            lazy[v<<1] = lazy[v];
19            lazy[(v<<1)|1] = lazy[v];
20        }
21        lazy[v].has_set = 0;
22        lazy[v].val = 0;
23    }
24
25    void push_sum(int v,int l,int r){
26        if(!lazy[v].has_sum)return;
27        tree[v].val = tree[v].val+lazy[v].val*(r-l+1);
28        if(l!=r){
29            lazy[v<<1].val = lazy[v<<1].val + lazy[v].val;
30            lazy[(v<<1)|1].val = lazy[(v<<1)|1].val + lazy[v].val;
31            if(!lazy[v<<1].has_set)lazy[v<<1].has_sum=1;
32            if(!lazy[(v<<1)|1].has_set)lazy[(v<<1)|1].has_sum=1;
33        }
34        lazy[v].has_sum = 0;
35        lazy[v].val= 0;
36    }
37
38    void update(int pos,ll val,int v,int tl, int tr){
39        if(tl == tr)tree[v] = {val};
40        else{
41            int tm = tl+(tr-tl)/2;

```

```

42        if(pos <= tm)update(pos,val,v<<1,tl,tm);
43        else update(pos,val,(v<<1)+1,tm+1,tr);
44        tree[v] = tree[v<<1]+tree[(v<<1)+1];
45    }
46 }
47
48 void update_set(int v,int tl,int tr,int l,int r,ll val){
49     push_set(v,tl,tr);
50     push_sum(v,tl,tr);
51     if(l>tr||r<tl)return;
52     if(tl>= l && tr <=r){
53         lazy[v].has_set=1;
54         lazy[v].has_sum=0;
55         lazy[v].val=val;
56         push_set(v,tl,tr);
57         return;
58     }
59     int tm = tl+(tr-tl)/2;
60     update_set(v<<1,tl,tm,l,r,val);
61     update_set((v<<1)|1,tm+1,tr,l,r,val);
62     tree[v] = tree[v<<1]+tree[(v<<1)|1];
63 }
64
65 void update_sum(int v,int tl,int tr,int l,int r,ll val){
66     push_set(v,tl,tr);
67     push_sum(v,tl,tr);
68     if(l>tr||r<tl)return;
69     if(tl>= l && tr <=r){
70         lazy[v].has_sum=1;
71         lazy[v].val += val;
72         push_sum(v,tl,tr);
73         return;
74     }
75     int tm = tl+(tr-tl)/2;
76     update_sum(v<<1,tl,tm,l,r,val);
77     update_sum((v<<1)|1,tm+1,tr,l,r,val);
78     tree[v] = tree[v<<1]+tree[(v<<1)|1];
79 }
80
81 Node query(int v ,int tl ,int tr ,int l ,int r){
82     push_set(v,tl,tr);
83     push_sum(v,tl,tr);
84     if(l>tr||r<tl)return {0};

```

```

85     if(tl>= l && tr <= r )
86         return tree[v];
87     int tm = tl+(tr-tl)/2;
88     return query(v<<1,tl,tm,l,r)+query((v<<1)|1,tm+1,tr,l,r);
89 }
90
91 };

```

2.7 Implicit Treap

```

1 struct node {
2     int val, sz, prior, lazy, sum, mx, mn, repl;
3     bool repl_flag, rev;
4     node *l, *r, *par;
5     node() {
6         lazy = 0;
7         rev = 0;
8         sum = 0;
9         val = 0;
10        sz = 0;
11        mx = 0;
12        mn = 0;
13        repl = 0;
14        repl_flag = 0;
15        prior = 0;
16        l = NULL;
17        r = NULL;
18        par = NULL;
19    }
20    node(int _val) {
21        val = _val;
22        sum = _val;
23        mx = _val;
24        mn = _val;
25        repl = 0;
26        repl_flag = 0;
27        rev = 0;
28        lazy = 0;
29        sz = 1;
30        prior = rnd();
31        l = NULL;
32        r = NULL;
33        par = NULL;

```

```

34    }
35    };
36
37 struct treap {
38     typedef node* pnode;
39     pnode root;
40     map<int, pnode> position; //positions of all the values
41     //clearing the treap
42     void clear() {
43         root = NULL;
44         position.clear();
45     }
46
47     treap() {
48         clear();
49     }
50
51     int size(pnode t) {
52         return t ? t->sz : 0;
53     }
54     void update_size(pnode &t) {
55         if(t) t->sz = size(t->l) + size(t->r) + 1;
56     }
57
58     void update_parent(pnode &t) {
59         if(!t) return;
60         if(t->l) t->l->par = t;
61         if(t->r) t->r->par = t;
62     }
63     //add operation
64     void lazy_sum_upd(pnode &t) {
65         if( !t or !t->lazy ) return;
66         t->sum += t->lazy * size(t);
67         t->val += t->lazy;
68         t->mx += t->lazy;
69         t->mn += t->lazy;
70         if( t->l ) t->l->lazy += t->lazy;
71         if( t->r ) t->r->lazy += t->lazy;
72         t->lazy = 0;
73     }
74     //replace update
75     void lazy_repl_upd(pnode &t) {
76         if( !t or !t->repl_flag ) return;

```

```

77     t->val = t->mx = t->mn = t->repl;
78     t->sum = t->val * size(t);
79     if( t->l ) {
80         t->l->repl = t->repl;
81         t->l->repl_flag = true;
82     }
83     if( t->r ) {
84         t->r->repl = t->repl;
85         t->r->repl_flag = true;
86     }
87     t->repl_flag = false;
88     t->repl = 0;
89 }
90 //reverse update
91 void lazy_rev_upd(pnode &t) {
92     if( !t or !t->rev ) return;
93     t->rev = false;
94     swap(t->l, t->r);
95     if( t->l ) t->l->rev ^= true;
96     if( t->r ) t->r->rev ^= true;
97 }
98 //reset the value of current node assuming it now
99 //represents a single element of the array
100 void reset(pnode &t) {
101     if(!t) return;
102     t->sum = t->val;
103     t->mx = t->val;
104     t->mn = t->val;
105 }
106 //combine node l and r to form t by updating corresponding queries
107 void combine(pnode &t, pnode l, pnode r) {
108     if(!l) {
109         t = r;
110         return;
111     }
112     if(!r) {
113         t = l;
114         return;
115     }
116     //Beware!!!Here t can be equal to l or r anytime
117     //i.e. t and (l or r) is representing same node
118     //so operation is needed to be done carefully
119     //e.g. if t and r are same then after t->sum=l->sum+r->sum operation

```

```

120     ,
121     //r->sum will be the same as t->sum
122     //so BE CAREFUL
123     t->sum = l->sum + r->sum;
124     t->mx = max(l->mx, r->mx);
125     t->mn = min(l->mn, r->mn);
126 }
127 //perform all operations
128 void operation(pnode &t) {
129     if( !t ) return;
130     reset(t);
131     lazy_rev_upd(t->l);
132     lazy_rev_upd(t->r);
133     lazy_repl_upd(t->l);
134     lazy_repl_upd(t->r);
135     lazy_sum_upd(t->l);
136     lazy_sum_upd(t->r);
137     combine(t, t->l, t);
138     combine(t, t, t->r);
139 }
140 //split node t in l and r by key k
141 //so first k+1 elements(0,1,2,...k) of the array from node t
142 //will be split in left node and rest will be in right node
143 void split(pnode t, pnode &l, pnode &r, int k, int add = 0) {
144     if(t == null) {
145         l = null;
146         r = null;
147         return;
148     }
149     lazy_rev_upd(t);
150     lazy_repl_upd(t);
151     lazy_sum_upd(t);
152     int idx = add + size(t->l);
153     if(t->l) t->l->par = null;
154     if(t->r) t->r->par = null;
155     if(idx <= k)
156         split(t->r, t->r, r, k, idx + 1), l = t;
157     else
158         split(t->l, l, t->l, k, add), r = t;
159     update_parent(t);
160     update_size(t);
161     operation(t);

```

```

162 }
163 //merge node l with r in t
164 void merge(pnode &t, pnode l, pnode r) {
165     lazy_rev_upd(l);
166     lazy_rev_upd(r);
167     lazy_repl_upd(l);
168     lazy_repl_upd(r);
169     lazy_sum_upd(l);
170     lazy_sum_upd(r);
171     if(!l) {
172         t = r;
173         return;
174     }
175     if(!r) {
176         t = l;
177         return;
178     }
179
180     if(l->prior > r->prior)
181         merge(l->r, l->r, r), t = l;
182     else
183         merge(r->l, l, r->l), t = r;
184
185     update_parent(t);
186     update_size(t);
187     operation(t);
188 }
189 //insert val in position a[pos]
190 //so all previous values from pos to last will be right shifted
191 void insert(int pos, int val) {
192     if(root == NULL) {
193         pnode to_add = new node(val);
194         root = to_add;
195         position[val] = root;
196         return;
197     }
198
199     pnode l, r, mid;
200     mid = new node(val);
201     position[val] = mid;
202
203     split(root, l, r, pos - 1);
204     merge(l, l, mid);

```

```

205     merge(root, l, r);
206
207     /*pnode x = new Node(val),merge(root,root,x);*/
208 }
209 //erase from qL to qR indexes
210 //so all previous indexes from qR+1 to last will be left shifted qR-qL
    +1 times
211 void erase(int qL, int qR) {
212     pnode l, r, mid;
213
214     split(root, l, r, qL - 1);
215     split(r, mid, r, qR - qL);
216     merge(root, l, r);
217 }
218 //returns answer for corresponding types of query
219 int query(int qL, int qR) {
220     pnode l, r, mid;
221
222     split(root, l, r, qL - 1);
223     split(r, mid, r, qR - qL);
224
225     int answer = mid->sum;
226     merge(r, mid, r);
227     merge(root, l, r);
228
229     return answer;
230 }
231 //add val in all the values from a[qL] to a[qR] positions
232 void update(int qL, int qR, int val) {
233     pnode l, r, mid;
234
235     split(root, l, r, qL - 1);
236     split(r, mid, r, qR - qL);
237     lazy_repl_upd(mid);
238     mid->lazy += val;
239
240     merge(r, mid, r);
241     merge(root, l, r);
242 }
243 //reverse all the values from qL to qR
244 void reverse(int qL, int qR) {
245     pnode l, r, mid;
246

```



```

247     split(root, l, r, qL - 1);
248     split(r, mid, r, qR - qL);
249
250     if(mid) mid->rev ^= 1;
251     merge(r, mid, r);
252     merge(root, l, r);
253 }
254 //replace all the values from a[qL] to a[qR] by v
255 void replace(int qL, int qR, int v) {
256     pnode l, r, mid;
257
258     split(root, l, r, qL - 1);
259     split(r, mid, r, qR - qL);
260     lazy_sum_upd(mid);
261     mid->repl_flag = 1;
262     mid->repl = v;
263     merge(r, mid, r);
264     merge(root, l, r);
265 }
266 //it will cyclic right shift the array k times
267 //so for k=1, a[qL]=a[qR] and all positions from qL+1 to qR will
268 //have values from previous a[qL] to a[qR-1]
269 //if you make left_shift=1 then it will to the opposite
270 void cyclic_shift(int qL, int qR, int k, bool left_shift = 0) {
271     if(qL == qR) return;
272     k %= (qR - qL + 1);
273
274     pnode l, r, mid, fh, sh;
275     split(root, l, r, qL - 1);
276     split(r, mid, r, qR - qL);
277
278     if(left_shift == 0) split(mid, fh, sh, (qR - qL + 1) - k - 1);
279     else split(mid, fh, sh, k - 1);
280     merge(mid, sh, fh);
281
282     merge(r, mid, r);
283     merge(root, l, r);
284 }
285 bool exist;
286 //returns index of node curr
287 int get_pos(pnode curr, pnode son = nullptr) {
288     if(exist == 0) return 0;
289     if(curr == NULL) {

```

```

290         exist = 0;
291         return 0;
292     }
293     if(!son) {
294         if(curr == root) return size(curr->l);
295         else return size(curr->l) + get_pos(curr->par, curr);
296     }
297
298     if(curr == root) {
299         if(son == curr->l) return 0;
300         else return size(curr->l) + 1;
301     }
302
303     if(curr->l == son) return get_pos(curr->par, curr);
304     else return get_pos(curr->par, curr) + size(curr->l) + 1;
305 }
306 //returns index of the value
307 //if the value has multiple positions then it will
308 //return the last index where it was added last time
309 //returns -1 if it doesn't exist in the array
310 int get_pos(int value) {
311     if(position.find(value) == position.end()) return -1;
312     exist = 1;
313     int x = get_pos(position[value]);
314     if(exist == 0) return -1;
315     else return x;
316 }
317 //returns value of index pos
318 int get_val(int pos) {
319     return query(pos, pos);
320 }
321 //returns size of the treap
322 int size() {
323     return size(root);
324 }
325 //inorder traversal to get indexes chronologically
326 void inorder(pnode cur) {
327     if(cur == NULL) return;
328     operation(cur);
329     inorder(cur->l);
330     cout << cur->val << '␣';
331     inorder(cur->r);
332 }

```

```

333
334 bool find(int val) {
335     if(get_pos(val) == -1) return 0;
336     else return 1;
337 }
338 };
339
340 treap t;
341 //Beware!!!here treap is 0-indexed
342
343 int main() {
344     int i, j, k, n, m, l, r, q;
345     for(i = 0; i < 10; i++) t.insert(i, i * 10);
346     t.cyclic_shift(4, 5, 1);
347     t.update(2, 5, 1);
348     t.replace(2, 5, 100);
349     t.reverse(2, 9);
350     t.replace(2, 5, 200);
351     cout << t.query(0, 7) << nl;
352     t.cyclic_shift(2, 3, 2, 1);
353     cout << t.get_pos(20) << nl;
354     t.erase(2, 2);
355     cout << t.find(30) << nl;
356     t.print_array();
357     return 0;
358 }

```

2.8 Trie

```

1 struct Node{
2     int v;
3     int cnt = 0;
4     Node* nxt[2];
5     Node(int __v){
6         v = __v;
7         cnt = 0;
8         nxt[0] = nxt[1] = NULL;
9     }
10 };
11
12 struct Trie{
13     typedef Node* pnode;
14     pnode root;

```

```

15     const int B = 30;
16
17 void insert(int val){
18     pnode cur = root;
19     for(int i=B;i>=0;i--){
20         bool b = val>>i&1;
21         if(cur->nxt[b]==NULL)cur->nxt[b] = new Node(b);
22         cur = cur->nxt[b];
23         cur->cnt++;
24     }
25 }
26
27 pnode erase(pnode cur, int val, int i) {
28     if (cur == NULL)return NULL;
29     cur->cnt--;
30     if (i < 0) {
31         if (cur->cnt == 0) {
32             delete cur;
33             return NULL;
34         }
35         return cur;
36     }
37
38     bool b = (val >> i) & 1;
39     cur->nxt[b] = erase(cur->nxt[b], val, i - 1);
40
41     if (cur->cnt == 0) {
42         delete cur;
43         return NULL;
44     }
45
46     return cur;
47 }
48
49 int query(int val){
50     pnode cur = root;
51     int ans = 0;
52     for(int i = B;i>=0;i--){
53         bool b = val>>i&1;
54         if(cur->nxt[!b])cur = cur->nxt[!b],ans<=&1,ans++;
55         else cur = cur->nxt[b],ans<=&1;
56     }
57     return ans;

```

```
58 }
59 };
```

3 Graphs

3.1 DFS

```
1 void dfs(int u){
2     vis[u] = 1;
3     for(auto v:adj[u]){
4         if(!vis[v])dfs(v);
5     }
6 }
```

3.2 BFS

```
1 queue<int> q;
2 q.push(0);
3 while(!q.empty()){
4     int u = q.front();
5     q.pop();
6     for(int v:adj[u]){
7         if (!vis[v]) {
8             vis[v] = 1;
9             q.push(v);
10        }
11    }
12 }
```

3.3 Topological Sort

```
1 vi ts;
2 queue<int> q;
3 for (int i = 1; i < n+1; i++)
4     if(!in[i])q.push(i);
5
6 while (!q.empty()) {
7     int u=q.front();
8     q.pop();
9     ts.pb(u);
10    for (int v : adj[u]) {
11        in[v]--;
12        if(!in[v])q.push(i);
13    }
```

```
14 }
15
16 if(sz(ts)!=n)cout <<"IMPOSSIBLE";
```

3.4 Flood Fill

```
1 int dx[] = {1,1,0,-1,-1,-1,0,1};
2 int dy[] = {0,1,1,1,0,-1,-1,-1};
3
4 int floodfill(int x,int y,char color1,char color2){
5     if(grid[x][y] == "outside")return 0;
6     if(grid[x][y] != color1)return 0;
7     int ans = 1;
8     grid[x][y] = color2;
9     for(int d=0;d<8;d++)ans += floodfill(x+dx[d],y+dy[d],color1,colr2);
10    return ans;
11 }
```

3.5 MST

3.5.1 Kruskal

```
1 struct Edge{
2     int u,v,w;
3     bool operator < (Edge b){return w<b.w;}
4 };
5
6 //inside main
7
8 vector<Edge> edges;
9 sort(all(edges));
10
11 DSU dsu(vertices);
12
13 for(int i=0;i<E;i++){
14     Edge front = edges[i];
15     if(!dsu.same(front.u,front.v)){
16         cost+= front.w;
17         dsu.merge(front.u,front.v);
18     }
19 }
```

3.5.2 Prims

```

1 vector<bool> vis(n);
2 priority_queue<pii, vector<pii>, greater<pii>> q;
3
4 auto process = [&](int u){
5     vis[u] = 1;
6     for(int i=0;i<sz(adj[u]);i++){
7         pii v = adj[u][i];
8         swap(v.S,v.F);
9         q.push(v);
10    }
11 }
12 process(0);
13
14 while(!q.empty()){
15     auto [w,u] = q.top();q.pop();
16     if(!vis[u]){cost += w;process(u);}
17 }

```

3.6 Dijkstra

3.6.1 Implementation

```

1 vll dist(n,INF64);
2 vector<bool> vis(n,0);
3 priority_queue<pll, vector<pll>, greater<pll>> pq;
4 pq.emplace(0,0);
5 dist[0] = 0;
6 vll cnt(n,0);//number of shortest paths
7 while(!pq.empty()){
8     auto [d1,u] = pq.top();
9     pq.pop();
10    if(vis[u])continue;
11    vis[u] = 1;
12    for(auto [v,c]:adj[u]){
13        if(dist[u]+c < dist[v]){
14            dist[v] = dist[u]+c;
15            pq.emplace(dist[v],v);
16            cnt[v] = cnt[u];
17        }else if(dist[u]+c == dist[v]){cnt[v] += cnt[u] }%MOD;
18    }
19 }

```

3.6.2 Space-State graph

```

1 vector<vll> dist(n,vll(1001,INF64));
2 vector<vector<bool>> vis(n,vector<bool> (1001,0));
3 typedef array<ll,3> T;
4 priority_queue<T ,vector<T>,greater<T>> pq;
5 pq.push({0,0,state[0]});
6 dist[0][state[0]] = 0;
7 while(!pq.empty()){
8     auto [d,u,s] = pq.top();
9     pq.pop();
10    if(vis[u][s])continue;
11    vis[u][s] = 1;
12    for(auto [v,w]:adj[u]){
13        ll c = min(s,slow[v]);
14        //Nuevo estad cuando llegue a V
15        if(dist[u][s]+w*s<dist[v][c]){
16            dist[v][c] = dist[u][s]+w*s;
17            pq.push({dist[v][c],v,c});
18        }
19    }
20 }

```

3.7 Bellman-Ford

3.7.1 Path

```

1 vector<int> d(n, INF);
2 d[v] = 0;
3 vector<int> p(n, -1);
4
5 for (;) {
6     bool any = false;
7     for (Edge e : edges)
8         if (d[e.a] < INF)
9             if (d[e.b] > d[e.a] + e.cost) {
10                 d[e.b] = d[e.a] + e.cost;
11                 p[e.b] = e.a;
12                 any = true;
13             }
14     if (!any)
15         break;
16 }
17 }

```

```

18 if (d[t] == INF)
19     cout << "No_path_from_" << v << "_to_" << t << ".";
20 else {
21     vector<int> path;
22     for (int cur = t; cur != -1; cur = p[cur]) path.push_back(cur);
23     reverse(path.begin(), path.end());
24 }

```

3.7.2 Cycle

```

1 vector<int> d(n, INF);
2 d[v] = 0;
3 vector<int> p(n, -1);
4 int x;
5 for (int i = 0; i < n; ++i) {
6     x = -1;
7     for (Edge e : edges)
8         if (d[e.a] < INF)
9             if (d[e.b] > d[e.a] + e.cost) {
10                 d[e.b] = max(-INF, d[e.a] + e.cost);
11                 p[e.b] = e.a;
12                 x = e.b;
13             }
14 }
15 if (x == -1) cout << "No_negative_cycle_from_" << v;
16 else{
17     int y = x;
18     for (int i = 0; i < n; ++i) y = p[y];
19     vector<int> path;
20     for (int cur = y;; cur = p[cur]) {
21         path.push_back(cur);
22         if (cur == y && path.size() > 1) break;
23     }
24     reverse(path.begin(), path.end());
25 }

```

3.8 Floyd-Warshall

```

1 for (int i = 0; i < n; ++i) d[i][i] = 0;
2
3 for (int k = 0; k < n; ++k)
4     for (int i = 0; i < n; ++i)
5         for (int j = 0; j < n; ++j)
6             if (d[i][k] < INF && d[k][j] < INF)

```

```

7 | d[i][j] = min(d[i][j], d[i][k] + d[k][j]);

```

3.9 SCC Kosaraju

```

1 class SCC{
2 private:
3     int n;
4     vector<vi> adj;
5     vector<vi> adj_t;
6     vector<vi> adj_cond;
7     vi order;
8     vector<bool> vis;
9     vi who;
10    int comp;
11 public:
12    SCC(int n,vector<vi> &a,vector<vi>&b):n(n),adj(a),adj_t(b){
13        comp = -1;
14        who.resize(n);
15        vis.resize(n);
16        for(int i=0;i<n;i++)
17            if(!vis[i]) dfs1(i);
18
19        vis.assign(n,0);
20        for(int i=n-1;i>=0;i--){
21            if(!vis[order[i]]){
22                comp++;
23                dfs2(order[i]);
24            }
25            adj_cond.resize(comp+1);
26            for(int u=0;u<n;u++)
27                for(auto v:adj[u])
28                    if(who[v] != who[u])
29                        adj_cond[who[u]].pb(who[v]);
30        }
31
32    void dfs1(int u) {
33        vis[u] = true;
34        for (int v:adj[u])
35            if (!vis[v]) dfs1(v);
36        order.pb(u);
37    }
38
39    void dfs2(int u) {

```

```

40     vis[u] = 1;
41     for (int v : adj_t[u])
42         if (!vis[v])dfs2(v);
43     who[u] = comp;
44 }
45 };

```

3.10 Flows

3.10.1 Dinic

```

1 struct flowEdge{
2     int u,v;
3     ll cap,flow = 0;
4     flowEdge(int u, int v, ll cap) : u(u), v(v), cap(cap) {};
5 };
6
7 struct Dinic{
8     vector<flowEdge> edges;
9     vector<vi> adj;
10    int n,s,t;
11    int id = 0;
12    vi level, next;
13
14    queue<int> q;
15
16    Dinic(int n, int s, int t) : n(n), s(s), t(t) {
17        adj.resize(n);
18        level.resize(n);
19        next.resize(n);
20        fill(all(level),-1);
21        level[s] = 0;
22        q.push(s);
23    }
24
25    void add(int u, int v, ll cap){
26        edges.emplace_back(u,v,cap);
27        edges.emplace_back(v,u,0);
28        adj[u].pb(id++);
29        adj[v].pb(id++);
30    }
31
32    bool bfs(){
33        while (!q.empty()){

```

```

34        int curr = q.front();
35        q.pop();
36        for (auto e : adj[curr]){
37            if (edges[e].cap - edges[e].flow < 1) continue;
38            if (level[edges[e].v] != -1) continue;
39            level[edges[e].v] = level[edges[e].u] + 1;
40            q.push(edges[e].v);
41        }
42    }
43    return level[t] != -1;
44 }
45
46 ll dfs(int u, ll flow){
47     if (flow == 0) return 0;
48     if (u == t) return flow;
49     for (auto& cid = next[u]; cid < adj[u].size(); cid++){
50         int e = adj[u][cid];
51         int v = edges[e].v;
52
53         if (level[edges[e].u] + 1 != level[v] || edges[e].cap - edges[e].
54             flow < 1) continue;
55         ll f = dfs(v,min(flow,edges[e].cap - edges[e].flow));
56         if (f == 0) continue;
57         edges[e].flow += f;
58         edges[e ^ 1].flow -= f;
59         return f;
60     }
61     return 0;
62 }
63
64 ll maxFlow(){
65     ll flow = 0;
66     while (bfs()){
67         fill(all(next),0);
68         for (ll f = dfs(s,INF64); f != 0ll; f = dfs(s,INF64)) flow += f;
69         fill(all(level),-1);
70         level[s] = 0;
71         q.push(s);
72     }
73     return flow;
74 }
75

```

```

76 vector<pii> minCut(){
77     vector<pii> ans;
78     fill(all(level),-1);
79     level[s] = 0;
80     q.push(s);
81     while (!q.empty()){
82         int curr = q.front();
83         q.pop();
84         for (int id = 0; id < adj[curr].size(); id++){
85             int e = adj[curr][id];
86             if (level[edges[e].v] == -1 && edges[e].cap - edges[e].flow > 0){
87                 q.push(edges[e].v);
88                 level[edges[e].v] = level[edges[e].u] + 1;
89             }
90         }
91     }
92
93     for (int i = 0; i < level.size(); i++){
94         if (level[i] != -1){
95             for (int id = 0; id < adj[i].size(); id++){
96                 int e = adj[i][id];
97                 if (level[edges[e].v] == -1 && edges[e].cap - edges[e].flow == 0)
98                     ans.emplace_back(edges[e].u, edges[e].v);
99             }
100         }
101     }
102     return ans;
103 }
104
105 vector<pii> MaximumMatching(){
106     vector<pii> ans;
107     fill(all(level),-1);
108     level[s] = 0;
109     q.push(s);
110     while (!q.empty()){
111         int curr = q.front();
112         q.pop();
113         for (int id = 0; id < adj[curr].size(); id++){
114             int e = adj[curr][id];
115             if (level[edges[e].v] == -1 && edges[e].cap - edges[e].flow == 0 &&
116                 edges[e].flow != 0ll){
117                 q.push(edges[e].v);
118                 level[edges[e].v] = level[edges[e].u] + 1;

```

```

118     }
119 }
120 }
121 for (int i = 0; i < level.size(); i++){
122     if (level[i] != -1)
123         for (int id = 0; id < adj[i].size(); id++){
124             int e = adj[i][id];
125             if (edges[e].u != s && edges[e].v != t && edges[e].cap - edges[e]
126                 ].flow == 0 && edges[e].flow != 0ll)
127                 ans.emplace_back(edges[e].u, edges[e].v);
128         }
129 }
130 return ans;
131 }
132 };

```

3.11 2-SAT

```

1 class SAT{
2 private:
3     int n;
4     vector<vi> adj;
5     vector<vi> adj_t;
6     vector<vi> adj_cond;
7     vi order;
8     vector<bool> vis;
9     vi who;
10    vector<bool> ans;
11    int comp;
12 public:
13
14    SAT(int m=0){
15        n = (m<<1);
16        adj.resize(n);
17        adj_t.resize(n);
18        vis.assign(n,0);
19        who.assign(n,-1);
20    }
21
22    void dfs1(int u) {
23        vis[u] = true;
24        for (int v:adj[u])

```

```

25     if (!vis[v])dfs1(v);
26     order.pb(u);
27 }
28
29 void dfs2(int u) {
30     vis[u] = 1;
31     for (int v : adj_t[u])
32         if (!vis[v])dfs2(v);
33     who[u] = comp;
34 }
35
36 //~a->b ~b->a
37 void addOR(int a, bool af, int b, bool bf) {
38     a += a + (af ^ 1);
39     b += b + (bf ^ 1);
40     adj[a ^ 1].push_back(b);
41     adj[b ^ 1].push_back(a);
42     adj_t[b].push_back(a ^ 1);
43     adj_t[a].push_back(b ^ 1);
44 }
45
46 //( a OR b ) ^ (~a or ~b)
47 void addXOR(int a, bool af, int b, bool bf) {
48     addOR(a, af, b, bf);
49     addOR(a, !af, b, !bf);
50 }
51 //(a -> b)
52 void _add(int a,bool af,int b,bool bf) {
53     a += a + (af ^ 1);
54     b += b + (bf ^ 1);
55     adj[a].push_back(b);
56     adj_t[b].push_back(a);
57 }
58
59 //(a->b)^(b->a)
60 void add(int a,bool af,int b,bool bf) {
61     _add(a, af, b, bf);
62     _add(b, !bf, a, !af);
63 }
64
65 bool solve_SAT(){
66     for(int i=0;i<n;i++)
67         if(!vis[i])dfs1(i);

```

```

68
69     comp = -1;
70     vis.assign(n,0);
71     for(int i=sz(order)-1;i>=0;i--)
72         if(!vis[order[i]]){
73             comp++;
74             dfs2(order[i]);
75         }
76     ans.assign(n/2,0);
77     for(int i=0;i<n;i+=2){
78         if(who[i] == who[i+1])return 0;
79         ans[i/2] = (who[i] > who[i+1]);
80     }
81     return 1;
82 }
83
84 bool operator[] (int i){
85     return ans[i];
86 }
87
88 };

```

3.12 Euler Paths and cycles

3.12.1 Directed Graphs

```

1 struct EulerSolver{
2 private:
3     vector<vector<pii>> adj;
4     int n,edges;
5     vi in,out;
6     vi ans,edge_ans;
7     int root;
8     vi done;
9     void dfs(int u){
10         while(done[u] < sz(adj[u])){
11             auto [v,id] = adj[u][done[u]++];
12             dfs(v);
13             edge_ans.pb(id);
14         }
15         ans.pb(u);
16     }
17 public:
18     EulerSolver(vector<vector<pii>> &a,int t):adj(a){

```



```

19  n = t;
20  edges = 0;
21  in.assign(n,0);
22  out.assign(n,0);
23  done.assign(n,0);
24  root = -1;
25  for(int u=0;u<n;u++){
26      for(auto v:adj[u]){
27          edges++;
28          in[v.F]++;
29          out[u]++;
30      }
31  }
32  }
33
34  bool possible(){
35      int cnt1=0,cnt2=0;
36      bool ok = 1;
37      for(int i=0;i<n;i++){
38          if (in[i] - out[i] == 1) cnt1++;
39          if (out[i] - in[i] == 1) cnt2++; root = i;
40          if (abs(in[i] - out[i]) > 1) ok = 0;
41      }
42      if(cnt1 > 1 || cnt2 > 1)ok = 0;
43      if(!ok)return 0;
44
45      if(root == -1)
46          for(int i=0;i<n;i++)
47              if(out[i])root = i;
48
49      if(root == -1)return 1;
50      dfs(root);
51
52      if(sz(ans) != edges+1)return 0;
53      reverse(all(ans));
54      reverse(all(edge_ans));
55      return 1;
56  }
57
58  int size(){
59      return ans.size();
60  }
61

```

```

62  int operator()(int i,bool x){
63      if(x)return ans[i];
64      else return edge_ans[i];
65  }
66  };

```

3.12.2 Unirected Graphs

```

1  struct EulerSolver{
2  private:
3      vector<vector<pii>> adj;
4      int n,edges;
5      vi ans,edge_ans;
6      vi deg;
7      int root;
8      vi done;
9      vector<bool> vis;
10
11  void dfs(int u){
12      while(done[u] < sz(adj[u])){
13          auto [v,id] = adj[u][done[u]++];
14          if(vis[id])continue;
15          vis[id]=1;
16          dfs(v);
17          edge_ans.pb(id);
18      }
19      ans.pb(u);
20  }
21  public:
22  EulerSolver(vector<vector<pii>> &a,int t):adj(a),n(t){
23      edges = 0;
24      done.assign(n,0);
25      deg.assign(n,0);
26      root = -1;
27      for(int u=0;u<n;u++){
28          for(auto v:adj[u]){
29              edges++;
30              deg[v.F]++;
31              deg[u]++;
32          }
33      vis.assign(edges/2,0);
34  }
35

```

```

36 bool possible(){
37     int odd = 0;
38     for(int i=0;i<n;i++){
39         if((deg[i]/2)&1){
40             odd++;
41             root = i;
42         }
43
44
45     if(odd>2)return 0;
46
47     if(root == -1)
48         for(int i=0;i<n;i++)
49             if(deg[i]){root = i;break;}
50
51     if(root == -1)return 1;
52     dfs(root);
53
54     if(sz(ans) != edges/2+1)return 0;
55     reverse(all(ans));
56     reverse(all(edge_ans));
57     return 1;
58 }
59
60 int size(){
61     return ans.size();
62 }
63
64 int operator()(int i,bool x){
65     if(x)return ans[i];
66     else return edge_ans[i];
67 }
68 };

```

3.13 Bridges

```

1 vector<pii> bridge;
2 vi disc(MAX),low(MAX);
3 vector<vi> adj(MAX);
4 int Time = 0;
5
6
7 void dfs(int u,int p){

```

```

8     disc[u] = low[u] = ++Time;
9     for(int v:adj[u]){
10         if(v==p)continue;
11         if(!disc[v]){
12             dfs(v,u);
13             // <= for articulation points
14             if(disc[u]<low[v])bridge.emplace_back(u,v);
15             low[u] = min(low[u],low[v]);
16         }else{
17             low[u] = min(low[u],disc[v]);
18         }
19     }
20 }
21 }

```

3.13.1 BridgeTree

```

1 //Bridges already found
2 vi who(n+1,-1);
3 int c = 1;
4
5 auto dfs2 = [&](auto &self,int u)->void{
6     who[u] = c;
7     for(auto [v,i]:adj[u]){
8         if(who[v]!=-1||bridge[i])continue;
9         self(self,v);
10    }
11 };
12
13
14 for(int i=1;i<=n;i++){
15     if(who[i]==-1)dfs2(dfs2,i),c++;
16 }
17
18 vector<vi> tree(c);
19
20 for(int i=0;i<m;i++){
21     if(bridge[i]){
22         tree[who[edges[i].u]].pb(who[edges[i].v]);
23         tree[who[edges[i].v]].pb(who[edges[i].u]);
24     }
25 }

```

3.14 Block Cut Tree

```

1 vector<vi> adj(n+1),bcc(n+1);
2 vi disc(n+1),low(n+1),who(n+1);
3 stack<int> st;
4 int Time = 0,sz = 0;
5
6 auto dfs = [&](auto &self,int u,int p)->void{
7     disc[u] = low[u] = ++Time;
8     st.push(u);
9     for(int v:adj[u]){
10         if(!disc[v]){
11             self(self,v,u);
12             low[u] = min(low[u],low[v]);
13             if(low[v] >= disc[u]){
14                 sz++;
15                 int x;
16                 do{
17                     x = st.top();
18                     st.pop();
19                     bcc[x].pb(sz);
20                 }while(x^v);
21                 bcc[u].pb(sz);
22             }
23             }else if(v!=p)low[u] = min(low[u],disc[v]);
24     }
25 };
26
27 for(int i=1;i<=n;i++)
28     if(!disc[i])dfs(dfs,i,0);
29
30 int c = sz+n+2;
31 vector<vi> tree(c);
32 vector<bool> art(c,0);
33 for(int i=1;i<=n;i++){
34     if(sz(bcc[i])>1){
35         who[i] = ++sz;
36         art[who[i]] = 1;
37         for(int v:bcc[i]){
38             tree[who[i]].pb(v);
39             tree[v].pb(who[i]);
40         }
41     }else if(sz(bcc[i])==1)who[i] = bcc[i][0];

```

42 | }

4 Trees

4.1 LCA - Binary Lifting

4.1.1 Normal LCA

```

1  const int MAX = 1e5+5, LG=18;
2  vi deep(MAX); // Si tiene peso vi cost(MAX)
3  int par[MAX][LG+1];
4
5  void dfs(int u = 1, int p=0) { // U = raiz del arbol
6      par[u][0] = p;
7      deep[u] = deep[p]+1; // cost[u] += cost[p]
8      for(int i=1; i<=LG; i++) par[u][i] = par[par[u][i-1]][i-1];
9      for(int v: adj[u]) {
10         if(v!=p) {
11             dfs(v, u);
12         }
13     }
14 }
15
16 int lca(int u, int v) {
17     if(deep[u] < deep[v]) swap(u, v);
18     for(int k=LG; k>=0; k--)
19         if(deep[par[u][k]] >= deep[v])
20             u = par[u][k];
21     if(u==v) return u;
22     for(int k=LG; k>=0; k--)
23         if(par[u][k] != par[v][k])
24             u = par[u][k], v = par[v][k];
25     return par[u][0];
26 }
27
28 int dist(int u, int v) {
29     int lc = lca(u, v);
30     return deep[u] + deep[v] - (deep[lc] << 1);
31 }
32
33 int kth(int u, int k) {
34     assert(k>=0);
35     for(int i=0; i<=LG; i++)
36         if(k && (1<<i)) u = par[u][i];
37     return u;

```

```

38 }
39
40 int isanc(int u, int g) {
41     int k = dep[u] - dep[g];
42     return k >= 0 && kth(u, k) == g;
43 }

```

4.1.2 Min/Max weight on the path

```

1  const int MAX = 1e5+5, LG=18;
2  vector<vector<pii>> adj(MAX, vector<pii> ());
3  vi deep(MAX);
4  int par[MAX][LG+1];
5  int mn[MAX][LG+1];
6  int mx[MAX][LG+1];
7
8  void dfs(int u=1, int p=0, int costmn = INF, int costmx = -INF) {
9      par[u][0] = p;
10     deep[u] = deep[p]+1;
11
12     if(p!=0) {
13         mn[u][0] = costmn;
14         mx[u][0] = costmx;
15     }
16     for(int i=1; i<=LG; i++) {
17         par[u][i] = par[par[u][i-1]][i-1];
18         mn[u][i] = min(mn[u][i-1], mn[par[u][i-1]][i-1]);
19         mx[u][i] = max(mx[u][i-1], mx[par[u][i-1]][i-1]);
20     }
21
22     for(auto [v, t]: adj[u]) {
23         if(v!=p) {
24             dfs(v, u, t, t);
25         }
26     }
27 }
28
29 pii minmax(int u, int v) {
30     pii ans = {INF, -INF};
31     if(deep[u] < deep[v]) swap(u, v);
32     for(int k=LG; k>=0; k--) if(deep[par[u][k]] >= deep[v]) {
33         ans.F = min(ans.F, mn[u][k]);
34         ans.S = max(ans.S, mx[u][k]);

```

```

35     u = par[u][k];
36 }
37 if(u==v)return ans;
38 for(int k=LG;k>=0;k--)if(par[u][k]!=par[v][k]){
39     ans.F = min({ans.F,mn[u][k],mn[v][k]});
40     ans.S = max({ans.S,mx[u][k],mx[v][k]});
41     u=par[u][k],v=par[v][k];
42 }
43 ans.F = min({ans.F,mn[u][0],mn[v][0]});
44 ans.S = max({ans.S,mx[u][0],mx[v][0]});
45
46 return ans;
47 }

```

4.1.3 Union path

```

1 pair<int, int> merge(pair<int, int> x, pair<int, int> y) {
2     if (x.first == 0)return y; //where {0,0} is a null path
3     if (y.first == 0)return x;
4     if (x.first == -1 || y.first == -1) return { -1, -1};
5     vector<int> can = {x.first, x.second, y.first, y.second};
6     int a = can[0];
7     for (int u : can)
8         if (dep[u] > dep[a])
9             a = u;
10    int b = -1;
11    for (int u : can)
12        if (!isanc(a, u)) {
13            if (b == -1) b = u;
14            if (dep[b] < dep[u]) b = u;
15        }
16    if (b == -1) {
17        b = can[0];
18        for (int u : can)
19            if (dep[u] < dep[b])
20                b = u;
21        return {a, b};
22    }
23    int g = lca(a, b);
24    for (int u : can) {
25        if (u == a || u == b)continue;
26        if (dep[u] < dep[g] || (!isanc(a, u) && !isanc(b, u)))return { -1,
-1};

```

```

27     }
28     return {a, b};
29 }

```

4.1.4 Intersection Path

```

1 pair<int, int> getCommonPath(int u, int a, int v, int b) {
2     // returns common path between u->a and v->b where u is ancestor a and v
    is ancestor b
3     if (!isanc(a, v)) return make_pair(0, 0);
4     int x = lca(a, b);
5     if (dep[v] < dep[u]) {
6         if (isanc(x, u))
7             return make_pair(u, x);
8     }else{
9         if (isanc(x, v))
10            return make_pair(v, x); // v->x is the common path
11    }
12    return make_pair(0, 0);
13 }
14 pair<int, int> path_union_light(pair<int, int> x, pair<int, int> y) {
15     if (x.first == 0) return y; //where {0,0} is a null path
16     if (y.first == 0) return x;
17     vector<int> can = {x.first, x.second, y.first, y.second};
18     int a = can[0];
19     for (int u : can) if (dep[u] > dep[a]) a = u;
20     int b = -1;
21     for (int u : can)if (!isanc(a, u)) {
22         if (b == -1) b = u;
23         if (dep[b] < dep[u]) b = u;
24     }
25     if (b == -1) {
26         b = can[0];
27         for (int u : can) if (dep[u] < dep[b]) b = u;
28         return {a, b};
29     }
30     return {a, b};
31 }
32 // returns the intersection of two paths (a,b) and (c,d)
33 // {0,0} for null path
34 pair<int, int> path_intersection(int a, int b, int c, int d) {
35     if (a == 0 || b == 0) return {0, 0};
36     int u = lca(a, b), v = lca(c, d); // splits both paths in two chains

```

```

    from lca to one node
37 pair<int, int> x, y;
38 x = getCommonPath(u, a, v, c);
39 y = getCommonPath(u, a, v, d);
40 pair<int, int> a1 = path_union_light(x, y);
41 x = getCommonPath(u, b, v, c);
42 y = getCommonPath(u, b, v, d);
43 pair<int, int> a2 = path_union_light(x, y);
44 return path_union_light(a1, a2);
45 }

```

4.2 Centroid Decomposition

```

1 class Centroid{
2 private:
3     vector<vi> tree;
4     vi par;
5     vi sub;
6     vector<bool> check;
7     int n;
8 public:
9     Centroid(vector<vi> &t):tree(t){
10         n = tree.size();
11         par.resize(n);
12         sub.resize(n);
13         check.resize(n);
14         build();
15     }
16
17     void build(int u = 0, int p = -1){
18         dfs(u, p);
19         int c = get_centroid(u,p,sub[u]);
20         check[c] = 1;
21         par[c] = p+1;
22
23         for (auto v : tree[c]){
24             if(!check[v])
25                 build(v, c);
26         }
27     }
28
29     int dfs(int u,int p){

```

```

31     if(check[u])return 0;
32     sub[u] = 1;
33     for(int v:tree[u])
34         if(v!=p)sub[u]+=dfs(v,u);
35
36     return sub[u];
37 }
38
39 int get_centroid(int u,int p,int x){
40     for(auto v:tree[u])
41         if(v!=p && sub[v]*2>x && !check[v])return get_centroid(v,u,x);
42
43     return u;
44 }
45
46 int operator[](int i){
47     return par[i];
48 }
49 };

```

4.3 Euler Tour Technique

```

1 int timer = 0;
2 vi start,final;
3 vector<vi> adj;
4 void dfs(int u = 0,int prev = -1){
5     start[u]=timer++;
6     for(int v:adj[u]){
7         if(v!=prev)dfs(v,u);
8     }
9     final[u] = timer;
10 }
11 // [start,final)

```

4.4 Tree Matching

```

1 vector<vll> dp(2,vll(MAX));
2
3 void dfs(ll u = 0, ll p=-1){
4     dp[0][u] = 0;
5     dp[1][u] = -INF;
6
7     for(ll v:adj[u]){
8         if(v == p)continue;

```

```

9     dfs(v,u);
10
11     dp[0][u] += max(dp[1][v], dp[0][v]);
12
13     ll opt = min(dp[0][v] - dp[1][v], 0ll);
14
15     dp[1][u] = max(dp[1][u], opt);
16 }
17
18 dp[1][u] += dp[0][u];
19 dp[1][u]++;
20 }

```

4.5 Diameter

```

1 vector<vector<pii>> adj;
2 vector<bool> vis;
3
4 int bfs(int n, vll &d, int v = 0){
5     vis.assign(n, 0);
6     d.assign(n, 0);
7     d[v] = 0;
8     queue<int> q;
9     q.push(v);
10    pll last = {v, 0};
11    vis[v] = 1;
12    while(!q.empty()){
13        int u = q.front(); q.pop();
14        for(auto [w, c]: adj[u]){
15            if(vis[w]) continue;
16            d[w] = d[u] + c;
17            q.push(w);
18            vis[w] = 1;
19            if(d[w] > last.S) last = {w, d[w]};
20        }
21    }
22    return int(last.F);
23 }
24
25 void solve(){
26     int n; cin >> n;
27     adj.assign(n, vector<pii> ());
28     for(int i=1; i<n; i++){

```

```

29     int a, b; ll c;
30     cin >> a >> b >> c;
31     adj[--a].pb({--b, c});
32     adj[b].pb({a, c});
33 }
34 vll dx(n), dy(n);
35 int a = bfs(n, dx);
36 int b = bfs(n, dy, a);
37 bfs(n, dx, b);
38
39 for(int i=0; i<n; i++){
40     cout << max(dx[i], dy[i]) << " \n" [i==(n-1)];
41 }
42
43 }

```

4.6 Heavy Light Decomposition

```

1 class HLD{
2 private:
3     int n;
4     vector<vi> adj;
5     vi parent;
6     vi heavy;
7     vi depth;
8     vi root;
9     vi treePos;
10    Segment_Tree tree;
11
12    int dfs(int u = 0){
13        int size = 1, mx_sub = 0;
14        for(auto v: adj[u]){
15            if(v != parent[u]){
16                parent[v] = u;
17                depth[v] = depth[u] + 1;
18                int sub = dfs(v);
19                if(sub > mx_sub){
20                    heavy[u] = v;
21                    mx_sub = sub;
22                }
23                size += sub;
24            }
25            return size;

```

```

26 }
27
28 template <class BinaryOperation>
29 void processPath(int u, int v, BinaryOperation op) {
30     for (; root[u] != root[v]; v = parent[root[v]]) {
31         if (depth[root[u]] > depth[root[v]]) swap(u, v);
32         op(treePos[root[v]], treePos[v] + 1);
33     }
34     if (depth[u] > depth[v]) swap(u, v);
35     op(treePos[u], treePos[v] + 1);
36 }
37
38 void init(vi &a){
39     for(int i=0,crt=0;i<n;++i){
40         if(parent[i] == -1 || heavy[parent[i]]!=i){
41             for(int j = i;j!=-1;j = heavy[j]){
42                 root[j] = i;
43                 treePos[j] = crt++;
44             }
45         }
46     }
47     tree = Segment_Tree(n);
48     for(int i=0;i<n;i++){
49         tree.update(treePos[i],a[i]);
50     }
51 }
52
53 public:
54
55 HLD(int n,vector<vi> &vale,vi &a):n(n),adj(vale){
56     parent.assign(n,0);
57     heavy.assign(n,-1);
58     depth.assign(n,0);
59     root.resize(n);
60     treePos.assign(n,0);
61     parent[0] = -1;
62     dfs();
63     init(a);
64 }
65
66 void update(int u,int &val){
67     tree.update(treePos[u],val);
68 }

```

```

69
70 void updatePath(int u, int v, int value) {
71     processPath(u, v, [this, &value](int l, int r) {
72         tree.update_range(l, r, value);
73     });
74 }
75
76 int query(int u,int v){
77     int ans = 0;
78     processPath(u,v,[this,&ans](int l,int r){
79         ans = max(ans,tree.query(l,r));
80     });
81     return ans;
82 }
83 };

```

4.7 Prufer Code

4.7.1 Build

```

1 auto dfs = [&](auto &self,int u){
2     for (int v:adj[u]) {
3         if (v != parent[u]) {
4             parent[v] = u;
5             dfs(v);
6         }
7     }
8 };
9
10 parent[n] = -1;
11 dfs(dfs,n);
12
13 int ptr = -1;
14 vi degree(n);
15 for(int i=1;i<=n;i++) {
16     degree[i] = sz(adj[i]);
17     if(degree[i]==1 && ptr==-1)
18         ptr = i;
19 }
20
21 vi code(n - 2);
22 int leaf = ptr;
23 for(int i=0;i<n-2;i++){
24     int next = parent[leaf];

```



```

25 code[i] = next;
26 if (--degree[next]==1 && next<ptr) {
27     leaf = next;
28 }else{
29     ptr++;
30     while (degree[ptr] != 1)
31         ptr++;
32     leaf = ptr;
33 }
34 }

```

4.7.2 Restore

```

1  int n = code.size() + 2;
2  vi degree(n,0);
3  for (int i : code)
4      degree[i]++;
5
6  int ptr = 0;
7  while (degree[ptr] != 1)
8      ptr++;
9
10 int leaf = ptr;
11
12 vector<pii> edges;
13 for(int v:code){
14     edges.emplace_back(leaf, v);
15     if (--degree[v] == 0 && v < ptr) {
16         leaf = v;
17     }else{
18         ptr++;
19         while (degree[ptr] != 1)
20             ptr++;
21         leaf = ptr;
22     }
23 }
24
25 edges.emplace_back(leaf, n-1);

```

4.7.3 Cayley's Formula

Cayley's formula states that the number of spanning trees in a complete labeled graph with n vertices is equal to:

$$n^{n-2}$$

In fact any Prufer code with $n - 2$ numbers from the interval $[0; n - 1]$ corresponds to some tree with n vertices. So we have n^{n-2} different such Prufer codes. Since each such tree is a spanning tree of a complete graph with n vertices, the number of such spanning trees is also n^{n-2} .

4.7.4 Number of ways to make a graph connected

$$s_1 \cdot s_2 \cdots s_k \cdot (s_1 + s_2 + \cdots + s_k)^{k-2} = s_1 \cdot s_2 \cdots s_k \cdot n^{k-2}$$

By accident this formula also holds for $k = 1$.

5 Math

5.1 FFT

```

1  struct base {
2      double a, b;
3      base(double a = 0, double b = 0) : a(a), b(b) {}
4      const base operator + (const base &c) const{
5          return base(a + c.a, b + c.b);
6      }
7      const base operator - (const base &c) const{
8          return base(a - c.a, b - c.b);
9      }
10     const base operator * (const base &c) const{
11         return base(a * c.a - b * c.b, a * c.b + b * c.a);
12     }
13 };
14 void fft(vector<base> &p, bool inv = 0) {
15     int n = p.size(), i = 0;
16     for(int j = 1; j < n - 1; ++j) {
17         for(int k = n >> 1; k > (i ^= k); k >>= 1);
18         if(j < i) swap(p[i], p[j]);
19     }
20
21     for(int l = 1, m; (m = l << 1) <= n; l <=<= 1) {
22         double ang = 2 * PI / m;
23         base wn = base(cos(ang), (inv ? 1. : -1.) * sin(ang)), w;
24         for(int i = 0, j, k; i < n; i += m) {
25             for(w = base(1, 0), j = i, k = i + 1; j < k; ++j, w = w * wn) {
26                 base t = w * p[j + 1];
27                 p[j + 1] = p[j] - t;
28                 p[j] = p[j] + t;

```

```

29     }
30 }
31 }
32
33 if(inv) for(int i = 0; i < n; ++i) p[i].a /= n, p[i].b /= n;
34 }
35 vector<ll> multiply(vi &a, vi &b) {
36     int n = a.size(), m = b.size(), t = n + m - 1, sz = 1;
37     while(sz < t) sz <= 1;
38     vector<base> x(sz), y(sz), z(sz);
39     for(int i = 0 ; i < sz; ++i) {
40         x[i] = i < (int)a.size() ? base(a[i], 0) : base(0, 0);
41         y[i] = i < (int)b.size() ? base(b[i], 0) : base(0, 0);
42     }
43     fft(x), fft(y);
44     for(int i = 0; i < sz; ++i) z[i] = x[i] * y[i];
45     fft(z, 1);
46     vector<ll> ret(sz);
47     for(int i = 0; i < sz; ++i) ret[i] = (long long) round(z[i].a);
48     while((int)ret.size() > 1 && ret.back() == 0) ret.pop_back();
49     return ret;
50 }

```

5.1.1 XOR

$$a + b = a \oplus b + 2(a \& b)$$

$$a + b = a|b + a \& b$$

$$a \oplus b = a|b - a \& b$$

$$1 \oplus 2 \oplus 3 \oplus \dots \oplus (4k - 1) = 0 \quad k \geq 0$$

5.2 Theorems

6 Strings

6.1 Hashing

6.1.1 Rolling Hashing

```

1 ll binpow(ll n,ll k,const int mod){
2     ll res = 1;
3     n%=mod;
4     while(k){

```

```

5         if(k&1)(res*=n)%=mod;
6         (n*=n)%=mod;
7         k>>=1;
8     }
9     return res;
10 }
11
12 const int MOD1 = 127657753,MOD2 = 987654319;
13 const int P1 = 137,P2 = 277;
14 const int MAX = 1e6+9;
15
16 vector<pll> P(MAX),I_P(MAX);
17
18 void calc(){
19     P[0] = {1,1};
20     for(int i=1;i<MAX;i++){
21         P[i].F = P[i-1].F*P1%MOD1;
22         P[i].S = P[i-1].S*P2%MOD2;
23     }
24     int i1 = binpow(P1,MOD1-2,MOD1);
25     int i2 = binpow(P2,MOD2-2,MOD2);
26     I_P[0] = {1,1};
27     for(int i=1;i<MAX;i++){
28         I_P[i].F = I_P[i-1].F*i1%MOD1;
29         I_P[i].S = I_P[i-1].S*i2%MOD2;
30     }
31 }
32
33 struct Hashing {
34     int n;
35     string s; // 0 - indexed
36     vector<pll> hs; // 1 - indexed
37
38     Hashing(string _s) {
39         n = _s.size();
40         s = _s;
41         hs.emplace_back(0, 0);
42         for (int i = 0; i < n; i++) {
43             pll p;
44             p.F = (hs[i].F + P[i].F * (s[i]-'a'+1) % MOD1) % MOD1;
45             p.S = (hs[i].S + P[i].S * (s[i]-'a'+1) % MOD2) % MOD2;
46             hs.push_back(p);
47         }

```

```

48 }
49
50 pll get_hash(int l, int r) { // 1 - indexed
51     pll ans;
52     ans.F = (hs[r].F - hs[l - 1].F + MOD1) * 1LL * I_P[l - 1].F % MOD1;
53     ans.S = (hs[r].S - hs[l - 1].S + MOD2) * 1LL * I_P[l - 1].S % MOD2;
54     return ans;
55 }
56 };

```

6.2 Suffix Array

```

1 struct suffix{
2     int index;
3     pii rank;
4     bool operator<(suffix b){return rank<b.rank;}
5     bool operator>(suffix b){return rank>b.rank;}
6 };
7
8 class Suffix_Array{
9     public:
10    string s;int n;
11    vector<suffix> suffixes;
12    vi lcp;
13    Suffix_Array(string x){
14        s = x+"$";
15        n = sz(s);
16        suffixes.resize(n);
17    }
18
19
20    void radix_sort(vector<suffix> &suff){
21        for(int i: vi{2,1}){
22            auto key = [&](const suffix &x){
23                return i == 1?x.rank.F:x.rank.S;
24            };
25
26            int mx = 0;
27            for (const auto &i:suff) { mx = max(mx, key(i)); }
28            vector<int> occs(mx + 1);
29            for (const auto &i:suff)occs[key(i)]++;
30            vector<int> start(mx + 1);
31            for (int i=1;i<=mx; i++) start[i] = start[i-1]+occs[i-1];

```

```

32
33    vector<suffix> new_arr(suff.size());
34    for (const auto &i : suff) {
35        new_arr[start[key(i)]] = i;
36        start[key(i)]++;
37    }
38    suff = new_arr;
39 }
40
41
42 void build(){
43     for(int i=0;i<n;i++)suffixes[i].index = i , suffixes[i].rank = {s[i]
44         ],(i+1<n?s[i+1]:-1)};
45     sort(all(suffixes));
46     vi equiv(n);
47
48     for(int i=1;i<n;i++){
49         auto c_val = suffixes[i].rank;
50         int cs = suffixes[i].index;
51         auto p_val = suffixes[i - 1].rank;
52         int ps = suffixes[i-1].index;
53         equiv[cs] = equiv[ps] + (c_val > p_val);
54     }
55
56     for(int cmp_ant = 1;cmp_ant <n; cmp_ant<=1){
57         for(auto &x:suffixes)
58             x.rank = {equiv[x.index],equiv[(x.index+cmp_ant)%n]};
59
60         //Std sort O(nlognlogn) Radix Sort O(nlogn)
61         radix_sort(suffixes);
62
63         for(int i=1;i<n;i++){
64             auto c_val = suffixes[i].rank;
65             int cs = suffixes[i].index;
66             auto p_val = suffixes[i - 1].rank;
67             int ps = suffixes[i-1].index;
68             equiv[cs] = equiv[ps] + (c_val > p_val);
69         }
70     }
71
72 void build_lcp(){
73     vector<int> suff_ind(n);

```

```

74   for (int i = 0; i < n; i++) suff_ind[suffixes[i].index] = i;
75   lcp.resize(n-1);
76   int start_at = 0;
77   for (int i = 0; i < n - 1; i++) {
78       int prev = suffixes[suff_ind[i] - 1].index;
79       int curr_cmp = start_at;
80       while (s[i + curr_cmp] == s[prev + curr_cmp]) curr_cmp++;
81       lcp[suff_ind[i] - 1] = curr_cmp;
82       start_at = max(curr_cmp - 1, 0);
83   }
84 }
85
86 void show(){
87     for(int i=0;i<n;i++)cout << suffixes[i].index << "␣\n"[(i+1)==n];
88 }
89 };

```

7 Others

7.1 Bnpow

```

1  ll binpow(ll a,ll b){
2      a%=MOD;
3      ll ans = 1;
4      while(b){
5          if(b&1)(ans*=a)%=MOD;
6          (a*=a)%=MOD;
7          b>>=1;
8      }
9      return ans;
10 }

```

7.2 Matrix exponentiation

```

1  struct Mat {
2      int n, m;
3      vector<vector<int>> a;
4      Mat() { }
5      Mat(int _n, int _m) {n = _n; m = _m; a.assign(n, vector<int>(m, 0)); }
6      Mat(vector< vector<int> > v) { n = v.size(); m = n ? v[0].size() : 0;
7          a = v; }
8      inline void make_unit() {
9          assert(n == m);

```

```

9      for (int i = 0; i < n; i++) {
10         for (int j = 0; j < n; j++) a[i][j] = i == j;
11     }
12 }
13 inline Mat operator + (const Mat &b) {
14     assert(n == b.n && m == b.m);
15     Mat ans = Mat(n, m);
16     for(int i = 0; i < n; i++) {
17         for(int j = 0; j < m; j++) {
18             ans.a[i][j] = (a[i][j] + b.a[i][j]) % mod;
19         }
20     }
21     return ans;
22 }
23 inline Mat operator - (const Mat &b) {
24     assert(n == b.n && m == b.m);
25     Mat ans = Mat(n, m);
26     for(int i = 0; i < n; i++) {
27         for(int j = 0; j < m; j++) {
28             ans.a[i][j] = (a[i][j] - b.a[i][j] + mod) % mod;
29         }
30     }
31     return ans;
32 }
33 inline Mat operator * (const Mat &b) {
34     assert(m == b.n);
35     Mat ans = Mat(n, b.m);
36     for(int i = 0; i < n; i++) {
37         for(int j = 0; j < b.m; j++) {
38             for(int k = 0; k < m; k++) {
39                 ans.a[i][j] = (ans.a[i][j] + 1LL * a[i][k] * b.a[k][j] % mod)
40                     % mod;
41             }
42         }
43     }
44     return ans;
45 }
46 inline Mat pow(long long k) {
47     assert(n == m);
48     Mat ans(n, n), t = a; ans.make_unit();
49     while (k) {
50         if (k & 1) ans = ans * t;
51         t = t * t;

```

```
51     k >>= 1;
52 }
53 return ans;
54 }
55 inline Mat& operator += (const Mat& b) { return *this = (*this) + b; }
56 inline Mat& operator -= (const Mat& b) { return *this = (*this) - b; }
57 inline Mat& operator *= (const Mat& b) { return *this = (*this) * b; }
58 inline bool operator == (const Mat& b) { return a == b.a; }
59 inline bool operator != (const Mat& b) { return a != b.a; }
60 };
61
62 // Usage
63 // Mat a(n, n);
64 // Mat b(n, n);
65 // Mat c = a * b;
66 // Mat d = a + b;
67 // Mat e = a - b;
68 // Mat f = a.pow(k);
69 // a.a[i][j] = x;
```

7.3 Random

```
1 | mt19937 rnd(chrono::steady_clock::now().time_since_epoch().count());
```